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Project: Azure Capacity Reservation Management Engine (ACRME)

Classification: Principal Cloud Architect — Architecture Governance

Version: 1.1

Date: August 2026

Status: Accepted

Part of: ACRME Architecture Decision Records — this is one of four standalone ADRs split from the consolidated ADR set.

About ADRs. An Architecture Decision Record captures a single significant architectural decision, the context that forced it, the options considered, the choice made, and its consequences. ADRs are immutable once accepted — a superseding decision is recorded as a new ADR rather than editing the original. Evidence tags: [Documented], [Decided], [Derived], [Assumed] (see Appendix).

ADR-003 — Capacity Management during Disaster Recovery (DR)

Status: Accepted

Date: August 2026

Deciders: Principal Cloud Architect, DR Owner, Platform Engineering, Security

Related decisions: D8, D10, D11; HC-1, HC-4, HC-6; G-14, G-15

Source: multi_region_placement_design.md §Emergency Capacity Transfer; acrme_production_readiness_review §29

Context

When a primary region fails, customers must recover in their DR region within a committed RTO. Pre-positioning a full 1:1 DR reserve is prohibitively expensive, yet under-provisioning risks a failed failover. During a crisis, waiting on Azure quota-increase approvals (hours) is unacceptable. The engine must also strictly separate **routine growth** from **destructive crisis operations** so that reconciliation can never accidentally trigger VM disruption.

Two capabilities are needed: - A way to reuse otherwise-idle NonProd capacity as DR overflow. - A safe, tiered escalation model for expanding DR capacity during a declared event.

Decision

Adopt **NonProd/DR co-location with a coverage floor**, a **formal two-mode engine state**, and a **three-tier emergency transfer model**:

1. **NonProd and DR may share a region (HC-1 constraint removed).** This lets the NonProd CRG's `effective_free` (after `dr_overflow_reserve`) count as DR overflow headroom. [Decided - D8]
 - **HC-6 DR_COVERAGE_FLOOR** guarantees the combined pool can absorb demand before DR placement is accepted:

$$\text{dr_crg_free_slots}(R) + \text{nonprod_crg_effective_free}(R) \geq \text{prod_vm_count} \times \text{dr_ratio_max}$$
 - **HC-4 DR_SEPARATION_CLASS** still guarantees Prod and DR sit in non-correlated failure domains (HIGH separation for non-paired regions).
2. **Two separate operating systems gated by engine_mode:** [Decided - D10]
 - **STEADY_STATE** — organic growth via the reconciliation loop and `CapacityIncreaseRequest` (ADR-004). Emergency Transfer is **rejected** in this mode.
 - **DR_EVENT_ACTIVE** — crisis operations only. The auto-increase trigger is **suppressed** in this mode. Mode transitions are operator-gated with dual approval (state machine `EngineModelState`, PRR §29), never automatic. [Decided - D10]
3. **Three-tier Emergency Capacity Transfer escalation:** [Decided - D11]
 - **Tier 1 — DirectExpansion (automated):** expand DR CR quantity using free headroom in the DR group. No approval beyond DR-event declaration.
 - **Tier 2 — QuotaNeutralTransfer (policy-gated):** reduce a NonProd CR (releasing quota to the shared NonProd+DR group pool) and expand the DR CR from that same pool. Net group headroom change = 0 — **quota-neutral, no VM execution-state change** (only NonProd SLA is removed). This is possible *only* because of the two-group model (ADR-002).
 - **Tier 3 — DestructiveTransfer (dual approval + elevated RBAC):** the only tier that modifies VM-to-CRG associations. The operator supplies an explicit `vm_disassociation_list` (no automated VM selection in Phase 1); executed via 6-step Path B with Path A fallback per VM. VMSS entries are rejected in Phase 1.
4. **Quota-neutral math (Tier 2):**

NonProd CR reduction → releases quota to shared NonProd+DR GROUP pool
 DR CR expansion → consumes from the SAME pool
 net group headroom change = 0 (ARM operations only; no Azure quota approval)
5. **DR reserve sizing** is 30–40% of Prod (`dr_ratio_min=0.30`, `dr_ratio_max=0.40`, `dr_ratio_target=0.35`); the protected floor uses `dr_ratio_max` (ADR-002/D7).
6. **Phase-1 safety posture:** Tier 2 is approval-gated; **Tier 3 is blocked** pending the G-14 consumer-credential model and G-15 engine-mode state machine. No invented SLA — propagation/approval times are measured and reported as unknown until observed.

Full Engine State Machine (normative) `engine_mode` is not a two-value flag but a **five-state machine** persisted in Cosmos DB with conditional writes, transition guards, and recovery tests — a production blocker until implemented (G-15): [Documented - §29]

State	Meaning	Permitted transitions
STEADY_STATE	Organic growth only; Emergency Transfer rejected	→ <code>DR_DECLARATION_PENDING</code>

State	Meaning	Permitted transitions
DR_DECLARATION_PENDING	Requested, awaiting dual approval + validation	→ DR_EVENT_ACTIVE (approved) · → STEADY_STATE (rejected/expired)
DR_EVENT_ACTIVE	Crisis operations only; auto-increase suppressed	→ FAILBACK_PENDING · → INCIDENT_HOLD
FAILBACK_PENDING	Failback requested, being validated	→ STEADY_STATE (completed) · → DR_EVENT_ACTIVE (validation failed) · → INCIDENT_HOLD
INCIDENT_HOLD	State conflict / critical failure — safe hold	→ DR_EVENT_ACTIVE · → FAILBACK_PENDING (on recovery approval)

All transitions are **operator-gated with dual approval** — never automatic.

EngineModeState Entity Must carry: environment/control-plane scope · current mode · state version · incident ID · requested-by · approved-by · transition timestamp · transition reason · active operation IDs · lease owner + expiry · recovery checkpoint. [Documented - §29]

DR Activation Semantics Entering DR_EVENT_ACTIVE **only establishes the operating mode** in which separately-governed emergency operations may be evaluated — it does **not** automatically authorize Tier 2 or Tier 3. Each tier is independently gated. The DR orchestrator validates group+subscription quota and CR/sharing state before starting any approved failover deployment, and records active-or-incident-hold state back to the state service. [Documented - §31]

Consequences

Positive: - Idle NonProd capacity doubles as DR overflow, cutting the cost of pre-positioned DR reserve while HC-6 guarantees sufficiency. - The **engine_mode** gate makes it structurally impossible for routine reconciliation to trigger destructive VM operations. - Tier 2 delivers meaningful crisis capacity **with zero VM disruption and zero Azure quota wait** — the common escalation path avoids Tier 3 entirely. - Every operation is tagged with its operating mode for a clean audit trail.

Negative / trade-offs: - Co-location adds risk that NonProd over-consumption erodes DR overflow; mitigated by HC-6, **dr_overflow_reserve**, and floor alerting. - **engine_mode** must be a formal Cosmos DB state propagated to every reconciliation cycle, with operator-gated, dual-approval transitions. - **Tier 3 is blocked** until G-14 (Managed Identity vs cross-tenant SP credential model) and G-15 (state machine) are resolved — a known Phase-1 limitation. - Tier 3 requires a rare, audited break-glass role (ACRME.SuperAdmin) for single-approver override. - Requires per-CRG-type **RegionalSnapshot** fields to evaluate HC-6.

Alternatives Considered

Alternative	Why Rejected
Keep DR_region NonProd_region	Prevents using NonProd as DR overflow — the whole point of co-location. [D8]
Remove separation with no compensating HC	Risks NonProd over-consumption leaving no DR headroom. [D8]
Unified capacity engine with mode flags	Mode flags create complex conditionals and risk triggering VM disassociation during routine reconciliation. [D10]
Emergency transfer as an extension of auto-increase	Conflates policy-driven growth with operator-gated crisis response; could run destructive ops without a DR declaration. [D10]
Two tiers only (automated + destructive)	Loses the quota-neutral Tier 2 — the critical low-risk intermediate that avoids Tier 3 in most crises. [D11]
Four tiers (separate VMSS tier)	VMSS disassociation deferred as a Phase-1 limitation rather than a distinct tier. [D11]

Appendix — Decision Log Cross-Reference

ADR	Primary Decisions	Hard Constraints	Key Gaps/Blockers
ADR-003 Capacity during DR	D8, D10, D11	HC-1, HC-4, HC-6	G-14 (credential), G-15 (engine mode), B-2 (POC-31)

Appendix — Status Legend

Status	Meaning
Proposed	Under discussion; not yet ratified
Accepted	Ratified and in force
Deprecated	No longer recommended but not yet replaced
Superseded	Replaced by a later ADR (referenced explicitly)

Appendix — Evidence Tag Taxonomy

Tag	Meaning
[Documented]	Traceable to Azure platform documentation or a formal FR/NFR
[Decided]	Explicit design choice in the Decision Log (D1–D11)
[Derived]	Logical consequence of a documented constraint or decision

Tag	Meaning
[Assumed]	Architectural judgement pending POC validation

Related ADRs

- **ADR-001 — Region Selection** (acrme_adr_001_region_selection.md)
- **ADR-002 — Quota and Capacity Management** (acrme_adr_002_quota_and_capacity_management.md)
- **ADR-004 — Forecast and Increase of Capacity and Quota** (acrme_adr_004_forecast_and_increase_o

Document Status: Accepted

Next Review: After POC-30 (Quota Groups GA) and POC-31 (quota release latency), and on resolution of G-14 / G-15.